

# Methodology: GPU DNS of Fractal-Interface Mixing in a Developing Duct

TorChannel — MERGE fractal-mixing study

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## Abstract

We describe the numerical methodology used to test whether a fractal (Koch) scalar interface enhances mixing in a laminar developing pipe flow, and how the Schmidt number  $Sc$  is swept to probe the high-Peclet regime. The pipeline couples a GPU-accelerated, double-precision direct numerical simulation (DNS) of the incompressible Navier–Stokes equations — staggered grid, fractional-step pressure projection, IMEX time integration, Brinkman-penalized immersed pipe wall — to a conservative passive-scalar transport solver. To make the high- $Sc$ , high-fractal-generation sweep tractable we exploit three physical properties of this configuration (parallel fully-developed flow, high-Peclet parabolicity, and an  $N$ -independent wall) that collapse the cost from  $O(Pe)$  pseudo-time steps per case to a single downstream sweep on a frozen, once-computed velocity field.

## 1 Problem setup

The configuration is a short, developing, laminar duct of square outer cross-section  $L_y \times L_z = 1 \times 1$  and streamwise length  $L_x$ , inside which a circular pipe of radius  $R = 0.42$  is imposed as an immersed solid. Two co-flowing streams of a passive scalar  $c \in [0, 1]$  enter the pipe at the inlet, separated by an interface. In the *baffle* configuration the dividing interface is a *Koch fractal* of generation  $N$  prescribed at the inlet; the pipe wall itself remains smooth and circular. The question is whether increasing the interfacial generation  $N$  — which multiplies the interfacial *area* without changing the 50/50 volume split — accelerates the downstream homogenisation of  $c$ , and how that effect scales with the Schmidt number

$$Sc = \frac{\nu}{D}, \quad Pe = Re \, Sc = \frac{U_b L}{D}, \quad (1)$$

where  $\nu$  is the kinematic viscosity,  $D$  the scalar diffusivity,  $U_b$  the bulk velocity and  $L$  a reference length. The molecular regime of interest (aqueous solutes) is  $Sc \sim 10^3$ , i.e.  $Pe$  up to  $\sim 10^4$  at the chosen  $Re = 40$ .

## 2 Governing equations

The flow obeys the incompressible Navier–Stokes equations, with the passive scalar advected by the resulting divergence-free velocity:

$$\nabla \cdot \mathbf{u} = 0, \quad (2)$$

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f}_{\text{ib}}, \quad (3)$$

$$\frac{\partial c}{\partial t} + \mathbf{u} \cdot \nabla c = D \nabla^2 c, \quad D = \frac{\nu}{Sc}. \quad (4)$$

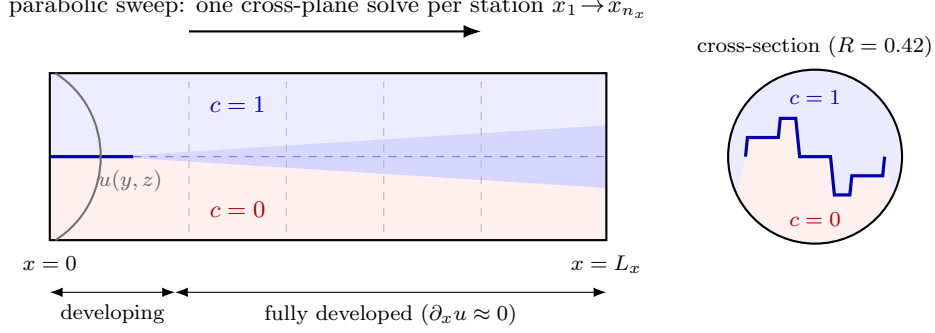


Figure 1: **(Sketch, schematic.)** Configuration. Two co-flowing scalar streams enter an immersed circular pipe separated by a Koch-folded interface (inset, cross-section). The fully-developed velocity is frozen and the steady scalar is obtained by a single downstream parabolic sweep.

All fields are stored in double precision throughout. The scalar is *passive*: it does not feed back into the momentum equation, so (4) carries no projection step.  $\mathbf{f}_{\text{ib}}$  is the immersed-boundary penalization force (Section 5). Because the conservative flux form  $\nabla \cdot (\mathbf{u}c)$  equals  $\mathbf{u} \cdot \nabla c$  for a divergence-free field, advection is discretised in flux form to conserve  $\langle c \rangle$  exactly.

### 3 Spatial discretisation

**Staggered MAC grid.** Velocities live on cell faces (the streamwise  $u$  on  $x$ -faces,  $v$  on  $y$ -faces,  $w$  on  $z$ -faces) while pressure and the scalar  $c$  are cell-centred. This Marker-and-Cell arrangement avoids odd–even pressure decoupling and gives a compact, exactly conservative divergence/gradient pair. All fields carry a single layer of ghost cells (indices 0 and  $-1$  in each direction) through which boundary conditions are imposed.

**Grid.** The mesh is uniform in the periodic-capable  $x$  and  $y$  directions ( $dx = L_x/n_x$ ,  $dy = L_y/n_y$ ) and *stretched* in the wall-normal  $z$  via a hyperbolic-tangent mapping, clustering points near the walls; the variable spacings  $dz_c$  (centre-to-centre) and  $dz_f$  (face-to-face) enter every  $z$ -derivative. Spatial operators are second-order: central differences for diffusion, and conservative flux differences for advection. Performance-critical kernels are fused for the GPU.

### 4 Time integration: fractional-step projection

Momentum is advanced by a semi-implicit (IMEX) fractional-step (projection) method. Within one step of size  $\Delta t$ :

1. **Explicit terms (Adams–Bashforth 2).** Advection and the in-plane  $(x, y)$  diffusion are evaluated explicitly and combined with second-order Adams–Bashforth,

$$\mathbf{u}^* = \mathbf{u}^n + \Delta t \left( \frac{3}{2} \mathbf{R}^n - \frac{1}{2} \mathbf{R}^{n-1} \right), \quad \mathbf{R} = \nu \nabla_{xy}^2 \mathbf{u} - (\mathbf{u} \cdot \nabla) \mathbf{u} + \mathbf{f}, \quad (5)$$

(the first step falls back to forward Euler).

2. **Implicit wall-normal diffusion ( $\theta$ -method).** The stiff  $z$ -diffusion is treated implicitly,  $(I - \theta \Delta t \nu \partial_{zz}) \mathbf{u}^{**} = \mathbf{u}^* + (1 - \theta) \Delta t \nu \partial_{zz} \mathbf{u}^*$ , with  $\theta = \frac{1}{2}$  (Crank–Nicolson). On the stretched grid this is a tridiagonal system per wall-normal column, solved by a batched Thomas solver over all columns at once. Treating only the  $z$ -diffusion implicitly removes the most restrictive viscous stability limit while keeping the linear solve one-dimensional.

3. **Immersed-boundary penalization** (Section 5) is applied to  $\mathbf{u}^{**}$  before projection.

4. **Pressure projection.** A Poisson equation enforces incompressibility,

$$\nabla^2 p = \frac{1}{\Delta t} \nabla \cdot \mathbf{u}^{**}, \quad \mathbf{u}^{n+1} = \mathbf{u}^{**} - \Delta t \nabla p, \quad (6)$$

solved by an FFT-based fast Poisson solver in the periodic  $x, y$  planes combined with a tridiagonal solve in  $z$  (a direct banded solver is available as a fallback). The resulting  $\mathbf{u}^{n+1}$  is discretely divergence-free.

## 5 Immersed pipe wall (Brinkman penalization)

The circular pipe is embedded in the Cartesian box by an implicit Brinkman (volume-penalization) force that drives the velocity to zero inside the solid:

$$\mathbf{f}_{\text{ib}} = -\frac{\chi}{\eta} \mathbf{u}, \quad \chi(\mathbf{x}) = \begin{cases} 1 & \text{solid (outside the pipe)} \\ 0 & \text{fluid (inside the pipe)} \end{cases} \quad (7)$$

with a small permeability  $\eta = 10^{-4}$ . The penalization is applied implicitly to the intermediate velocity *before* the projection,  $\mathbf{u} \leftarrow \mathbf{u}/(1 + \Delta t \chi/\eta)$ , so the fast FFT-Poisson solver is left untouched. The solid mask  $\chi$  is sampled separately on each staggered face ( $\chi_u, \chi_v, \chi_w$ ) and at cell centres ( $\chi_c$ ); here it is the complement of the inscribed disc of radius  $R = 0.42$  centred on the cross-section, giving a solid fraction  $\approx 0.45$ . The bulk-flux controller and all mixing diagnostics are restricted to fluid cells ( $\chi_c < \frac{1}{2}$ ) so that scalar diffusing into the (non-physical) solid is never counted as mixing.

## 6 Inflow / outflow boundary conditions

The duct is open (`bc_x = inout`): periodic in  $y$  only where specified, walls otherwise, and a prescribed inlet / convective outlet in  $x$ . The inlet streamwise velocity is a smooth duct profile, the product of parabolae in  $y$  and  $z$  that vanish at the walls,

$$u_{\text{in}}(y, z) \propto \frac{4y(L_y - y)}{L_y^2} \frac{4z(L_z - z)}{L_z^2}, \quad (8)$$

masked to zero inside the solid and normalised so the fluid-area-mean inlet velocity equals  $U_b$ . The outflow imposes zero streamwise gradient with a global mass-flux correction enforcing  $\int u_{\text{out}} = \int u_{\text{in}}$  over the fluid face, so the discrete domain conserves mass. The scalar inlet ghost is pinned to the prescribed interface  $c_{\text{inlet}}(y, z)$  each step; the scalar outflow is zero-gradient. Lateral walls use no-flux ( $\partial_n c = 0$ ) for the scalar.

## 7 The baffle: a Koch-fractal inlet interface

In the *baffle* mode the pipe wall is smooth; the only  $N$ -dependent ingredient is the *scalar interface* prescribed at the inlet. The two streams meet on a curve that runs wall-to-wall across the pipe and is Koch-folded in the  $y$ -direction at generation  $N$ .

**Generator.** The interface is built from an area-balanced four-segment zigzag motif, each segment of length  $1/r$ :

$$(0, 0) \rightarrow (\tfrac{1}{4}, a) \rightarrow (\tfrac{1}{2}, 0) \rightarrow (\tfrac{3}{4}, -a) \rightarrow (1, 0), \quad a = \sqrt{(1/r)^2 - (1/4)^2},$$

valid for  $1 < r < 4$ . Each outward triangle is matched by an equal inward indentation, so the fold preserves the exact 50/50 area split at every generation (the mean scalar is held at  $\langle c \rangle = \frac{1}{2}$ ). Replacing every edge by an affine copy of the motif  $N$  times yields a generation- $N$  polyline with fractal dimension

$$D_f = \frac{\log 4}{\log r}. \quad (9)$$

The interfacial length (hence the area available for diffusive flux) grows as  $\sim (4/r)^N$  relative to the flat  $N = 0$  baseline, while the enclosed area is unchanged — isolating the effect of interfacial area on mixing.

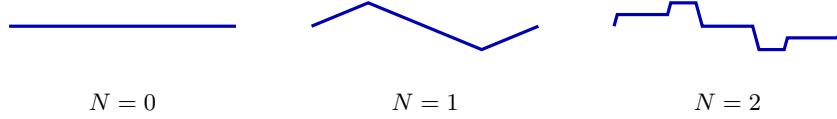


Figure 2: Area-balanced Koch generator ( $r = 3$ ,  $D_f = \log 4 / \log 3 \approx 1.26$ ). Each edge is replaced by the four-segment motif, multiplying interfacial length by  $\sim 4/r$  per generation while preserving the 50/50 split. The inlet interface uses generation  $N$  of this fold; these are the actual polylines imposed.

**Rasterisation.** The cross-sectional field  $c(y, z)$  is set from the signed distance  $\phi$  to the open Koch polyline, smoothed over a prescribed interface thickness  $\varepsilon$  (a fixed number of cells):

$$c(y, z) = \frac{1}{2} [1 + \tanh(\phi/\varepsilon)]. \quad (10)$$

The distance *magnitude* comes from the open polyline, but its *sign* (which stream a cell belongs to) is taken from an even-odd point-in-polygon test against the interface closed along the far wall. This is robust at high  $N$ , where a naive nearest-segment side test produces wrong-sign corner artefacts ( $N \geq 3$ ). The interface is homogeneous in  $x$  at the inlet; all streamwise structure that develops downstream is produced by the flow. A companion *surface\_baffle* mode additionally folds the *pipe wall* into a Koch surface, differing from *baffle* only by the wall, for a controlled isolation of the surface-roughness effect.

## 8 Passive-scalar transport

The scalar lives at cell centres and is advanced with the same IMEX split as the momentum equations (AB2 for advection and in-plane diffusion;  $\theta$ -method for wall-normal diffusion), reusing the batched tridiagonal solver. Advection is in conservative flux form,

$$\nabla \cdot (\mathbf{u}c) = \frac{F_{i+\frac{1}{2}}^x - F_{i-\frac{1}{2}}^x}{dx} + \frac{F_{j+\frac{1}{2}}^y - F_{j-\frac{1}{2}}^y}{dy} + \frac{F_{k+\frac{1}{2}}^z - F_{k-\frac{1}{2}}^z}{dz_f}, \quad (11)$$

with face fluxes  $F = U_{\text{face}} c_{\text{face}}$  built from the MAC face velocities. Two face reconstructions are available:

- **Central** ( $c_{\text{face}}$  = linear average) — second-order, used at moderate cell-Peclet.
- **van Leer TVD** — a MUSCL upwind reconstruction with the van Leer limiter,  $c_{\text{face}} = c_{\text{up}} + \frac{1}{2} \psi$ , monotone (no dispersive over/undershoot) at the high cell-Peclet of high- $Sc$  runs. Since the face velocity is identically zero at every wall, the limiter never alters a wall flux.

The wall-normal boundary condition is no-flux (Neumann,  $\partial_z c = 0$ ), which conserves the scalar mean and lets the segregation variance decay to zero — the correct closure for a mixing / decay study. A Dirichlet ( $c = 0$ ) option exists for future sustained-gradient setups.

## 9 Mixing diagnostics

Mixing is quantified by the *intensity of segregation*, the fluid-volume-weighted, variance-based mixedness

$$M = \frac{\sigma_c}{\sigma_{\max}}, \quad \sigma_c^2 = \langle (c - \bar{c})^2 \rangle, \quad \sigma_{\max} = \sqrt{\bar{c}(1 - \bar{c})}, \quad (12)$$

where  $\langle \cdot \rangle$  is the  $z$ -stretch-weighted average over fluid cells only.  $M = 1$  is fully segregated (two pure streams);  $M \rightarrow 0$  is fully mixed. For the developing duct the relevant object is the per-slice profile  $M(x)$ , evaluated plane-by-plane, from which a *mixing length*  $L_{\text{mix}}$  is read off as the streamwise station where  $M(x)$  first falls below a threshold (e.g.  $M = 0.9$ ), linearly interpolated between planes. The fractal effect is reported as the ratio  $L_{\text{mix}}(N)/L_{\text{mix}}(0)$ . A complementary scalar-dissipation diagnostic  $\langle |\nabla c|^2 \rangle$  is also available; unlike the gravest-mode-dominated variance, it is directly sensitive to interfacial area and so captures the fractal's contribution that the variance metric alone can miss.

## 10 Acceleration: from $O(Pe)$ to one sweep

A direct time-march of (4) to steady state costs  $\sim Pe$  pseudo-time steps and, at  $Sc = 10^3$  on a fine cross-section, is infeasible. Three properties of this configuration collapse the cost.

### 10.1 Parabolic space-marching (high-Peclet) scalar solver

In a developing duct with  $u > 0$  everywhere (no streamwise recirculation), the steady scalar balance

$$u \frac{\partial c}{\partial x} = D \left( \frac{\partial^2 c}{\partial y^2} + \frac{\partial^2 c}{\partial z^2} \right) - \left( v \frac{\partial c}{\partial y} + w \frac{\partial c}{\partial z} \right) \quad (13)$$

is obtained after *dropping the streamwise diffusion*  $\partial_{xx}c$  — the *parabolic* (boundary-layer) approximation, asymptotically exact as  $Pe \rightarrow \infty$  and, unlike a time-march, free of streamwise *numerical* diffusion. The equation is then parabolic in  $x$  and the steady field follows from a *single downstream sweep* — one cross-plane solve per  $x$ -station — instead of marching the unsteady field to steady state. The streamwise term uses second-order upwind (BDF2) from the two already-solved upstream planes; each cross-plane  $(y, z)$  diffusion solve uses alternating-direction implicit (ADI) line relaxation (a  $z$ -implicit then a  $y$ -implicit tridiagonal sweep), which converges in a handful of inner iterations — fastest precisely at high  $Sc$ , where the streamwise term dominates the diagonal. Cost is  $O(n_x)$  plane solves rather than  $O(Pe)$  full-field updates. For the smooth baffle  $v = w = 0$ , so transverse advection is switched off entirely.

### 10.2 Frozen, fully-developed velocity (parallel-flow assumption)

The scalar marcher runs on a *frozen* velocity field. The pipe wall is straight and  $x$ -independent, so the laminar flow becomes *fully developed* a short distance past the inlet: beyond the entrance length the profile  $u(y, z)$  stops changing with  $x$  ( $\partial_x u \approx 0$ ) and every downstream plane is identical. We verified this directly on the computed field — the cross-sectional profile is within 0.1% of its developed shape by  $x \approx 2.8$  and  $\partial_x u \sim 5 \times 10^{-5}$  thereafter. Consequently the velocity need only be solved over the short developing region, and extending it to an arbitrarily long scalar domain is exact: the marcher edge-clamps the streamwise interpolation to the developed profile, i.e. it repeats the one converged  $u(y, z)$  downstream. This is the same parallel-flow premise that justifies dropping  $\partial_{xx}c$  in the first place; it holds because the baffle wall does not vary with  $x$  (it would *not* hold for a streamwise-varying cross-section).

### 10.3 $N$ -independent velocity, solved coarse and reused

In the baffle mode the Koch generation  $N$  lives only in the inlet *scalar*; the wall — hence the entire velocity field — is identical for every  $N$ . The developed velocity is therefore solved *once* and cached, then reused across the whole  $N$ -sweep. Moreover the smooth-pipe flow is well resolved on a *coarse* cross-section, whereas only the scalar needs the fine grid to resolve the Koch striations without false diffusion. We exploit this by solving the velocity on a coarse mesh and tri-linearly interpolating it (zeroed inside the fine solid mask) onto the fine scalar grid. The full  $N$ -sweep thus costs *one* coarse velocity solve plus  $N$  cheap parabolic scalar sweeps, in place of one fine velocity solve and a high- $Sc$  time-march *per*  $N$ .

## 11 The Schmidt-number sweep

A sweep fixes the geometry ( $R$ ,  $L_x$ , grid) and the threshold, and varies the Koch generation  $N = 0, 1, 2, \dots$  at a target Schmidt number. For each  $N$  the parabolic marcher returns the steady  $c(x, y, z)$  on the frozen developed velocity; from it we extract  $M(x)$ , the mid-plane and cross-section slices, and  $L_{\text{mix}}(N)$ . Comparing  $L_{\text{mix}}(N)/L_{\text{mix}}(0)$  across  $N$ , at increasing  $Sc$  (equivalently  $Pe = Re Sc$ ), tests whether the fractal interface’s extra area shortens the mixing length and whether that advantage persists into the molecular ( $Sc \sim 10^3$ ) regime.

**Validation.** The scalar solver is checked against an analytic error-function diffusion profile, exact mean conservation, and zero numerical streamwise diffusion; the DNS divergence is monitored every step and the run aborts if it grows.

## 12 Preliminary results ( $Sc = 10, 100$ and $1000$ )

Table 1 and Figure 3 collect the mixing length  $L_{\text{mix}}$  (the station where  $M(x)$  first crosses the threshold) for the baffle, at the three Schmidt numbers run. The  $Sc = 100$  and  $Sc = 1000$  data come from the parabolic marcher; the  $Sc = 10$  data from a direct time-march to steady state (an independent cross-check of the marcher). Because  $L_{\text{mix}}$  grows with  $Pe$ , the  $Sc = 1000$  sweep uses a much longer domain ( $L_x = 90$  vs. 6) so that all  $N$  cross  $M = 0.9$  within it; the threshold is therefore the *same*  $M = 0.9$  for every  $Sc$ , an apples-to-apples comparison. The ratio  $L_{\text{mix}}(N)/L_{\text{mix}}(0)$  *does* depend on the threshold — it rises smoothly toward 1 as  $M$  is lowered (Figure 4) — so every comparison here is made at a *fixed* threshold; at fixed  $M$  the  $N$ -ordering and trend are robust, and only the reported magnitude shifts with the choice of  $M$ .

|     | $Sc = 10$ ( $M=0.9$ ) |       | $Sc = 100$ ( $M=0.9$ ) |       | $Sc = 1000$ ( $M=0.9$ ) |       |
|-----|-----------------------|-------|------------------------|-------|-------------------------|-------|
| $N$ | $L_{\text{mix}}$      | ratio | $L_{\text{mix}}$       | ratio | $L_{\text{mix}}$        | ratio |
| 0   | 0.492                 | 1.000 | 5.392                  | 1.000 | 62.818                  | 1.000 |
| 1   | 0.303                 | 0.616 | 3.821                  | 0.709 | 38.370                  | 0.611 |
| 2   | 0.198                 | 0.402 | 2.358                  | 0.437 | 26.258                  | 0.418 |
| 3   | 0.152                 | 0.309 | 1.975                  | 0.366 | 22.164                  | 0.353 |
| 4   | 0.136                 | 0.276 | 1.914                  | 0.355 | 21.657                  | 0.345 |
| 5   | —                     | —     | —                      | —     | 21.581                  | 0.344 |

Table 1: Mixing length  $L_{\text{mix}}$  (station where  $M(x)$  crosses the threshold) and its ratio to the flat  $N = 0$  baseline, baffle mode,  $R = 0.42$ . All at the same threshold  $M = 0.9$ ;  $Sc = 10, 100$  use  $L_x = 6$ ,  $Sc = 1000$  uses  $L_x = 90$  (the longer domain the high- $Pe$  field needs to reach  $M = 0.9$ ).

The mechanism is visible directly in the cross-sections (Figure 5): higher  $N$  injects more interfacial length into the same disc, and the steep transverse scalar gradients it creates are

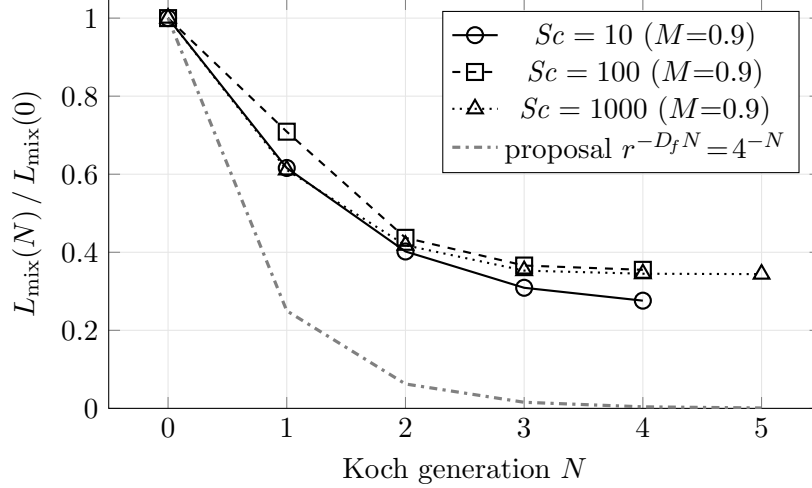


Figure 3: Fractal mixing enhancement, all at a fixed threshold  $M = 0.9$ : the mixing length falls with generation  $N$  and saturates near 0.35 by  $N \approx 3$ , *roughly independently of  $Sc$*  (the three curves nearly overlap;  $Sc = 100$  is marginally the weakest). The grey dash-dotted line is the proposal’s diffusive-limit prediction  $r^{-D_f N} = 4^{-N}$  (Eq. 4): the measured ratios fall well short of it — the finite- $Pe$  duct realises only part of the ideal exponential cut. The ratio’s magnitude depends on the threshold (Figure 4), but its  $N$ -ordering does not.

ground down by diffusion over a shorter streamwise distance. The companion  $M(x)$  decay (Figure 6) shows the same ordering as a single scalar per station, for both  $Sc = 100$  and  $Sc = 1000$ .

### 13 Implications

1. **The fractal interface robustly shortens the mixing length, roughly independently of  $Sc$ .** At a fixed threshold ( $M = 0.9$ ), folding to  $N = 3$  cuts  $L_{\text{mix}}$  to 0.31, 0.37 and 0.35 of the flat baseline at  $Sc = 10, 100, 1000$  — a  $\sim 3\times$  reduction that saturates near 0.35 for  $N \geq 3$  at every  $Sc$ , with the same volume split and the same wall. This confirms, in a resolved laminar duct, the core MERGE premise that interfacial *area* (not bulk stirring) drives the enhancement. (At a *common* threshold there is no clear monotone  $Sc$ -trend — the apparent strengthening with  $Sc$  in earlier mixed-threshold readings was a threshold artifact, see Fig. 4.)
2. **Diminishing returns.** Most of the gain is in the first folds ( $N: 0 \rightarrow 1 \rightarrow 2$ ); the ratio then flattens, saturating by  $N \approx 3-4$  at every  $Sc$  (e.g.  $Sc = 1000$  gives 0.42, 0.35, 0.345, 0.344 at  $N = 2, 3, 4, 5$ ). Folds finer than the local diffusion length are smeared away before they can laminate, so deeper fractals stop paying off.
3.  **$L_{\text{mix}}$  scales steeply with  $Pe$ .** The  $N = 0$  baseline grows  $0.49 \rightarrow 5.4 \rightarrow 63$  for  $Sc = 10 \rightarrow 100 \rightarrow 1000$  (all at  $M = 0.9$ ) — about a decade per decade of  $Sc$ . This is why a high- $Sc$  run needs a long domain, and is the direct motivation for the parabolic-marcher / frozen-velocity acceleration of Section 10.
4. **Interfacial folding dominates over wall roughness.** The *surface\_baffle* (Koch folds added to the pipe *wall*) gives a weaker reduction than the interface baffle at equal  $N$  (e.g. at  $Sc = 10, N = 3$ :  $L_{\text{mix}} = 0.21$  vs. 0.15), indicating that the dominant mechanism is lamination of the scalar interface, not corrugation-driven secondary flow.
5. **The acceleration is what makes the proposal regime reachable.** Because the

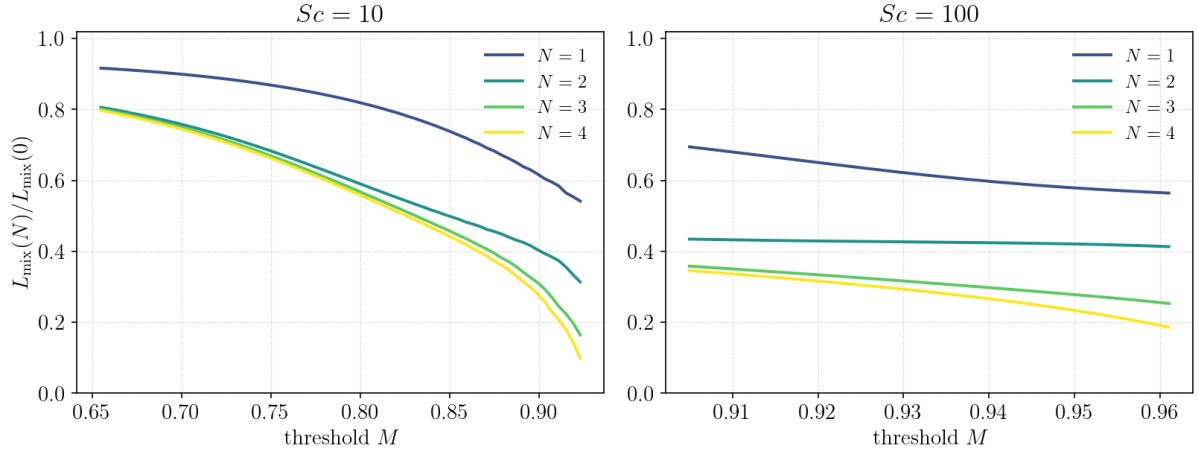


Figure 4: Threshold sensitivity of the ratio  $L_{\text{mix}}(N)/L_{\text{mix}}(0)$ : each curve is a generation  $N$  (colour as in Fig. 6) swept over the mixedness threshold  $M$ . The ratio rises smoothly and *monotonically* toward 1 as  $M$  is lowered — it never flickers or reverses — and the  $N$ -ordering is preserved at every  $M$ . The fractal advantage is therefore largest near the inlet (high  $M$ ) and fades toward complete mixing, but the qualitative conclusions are threshold-independent; only the reported magnitude depends on the chosen  $M$ . ( $Sc = 100$  spans only a narrow band here because its  $L_x = 6$  domain mixes to just  $M \approx 0.84$ ; a longer-domain run widens it to match the  $Sc = 10$  range.)

velocity is fully developed and  $N$ -independent, the entire multi- $N$ , high- $Sc$  sweep collapses to one coarse velocity solve plus cheap parabolic scalar sweeps. Without it, true  $Sc \sim 10^3$  by direct time-march would be infeasible; with it, the  $Sc = 10^3$  sweep runs in minutes per generation.

**Caveats.** All ratios are quoted at a common  $M = 0.9$ . The  $Sc = 10$  values come from a time-march on a coarse grid (some cases stopped at the step cap, not fully steady) while  $Sc = 100$  and  $Sc = 1000$  use the parabolic marcher on fine grids, so cross- $Sc$  comparisons are indicative rather than exact — in particular the  $Sc = 10$  ratios sit slightly below the marcher cases and should not be over-read as a trend. The configuration and generator diagrams (Figs. 1, 2) are schematic; the quantitative figures (Figs. 3, 4, 5, 6) are computed results.



Sc= 1000 baffle:  $c(y, z)$

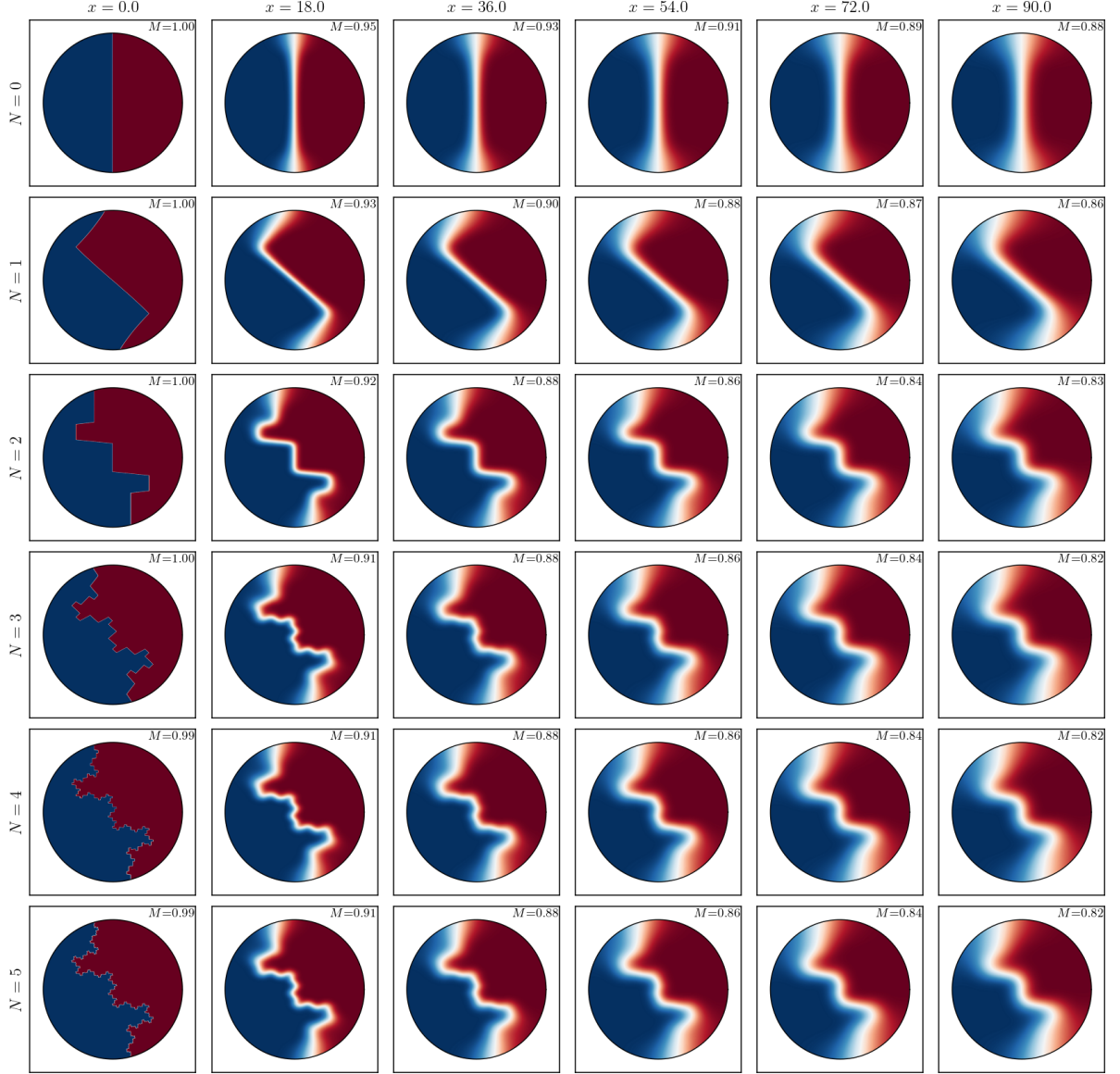


Figure 5: Steady scalar  $c(y, z)$  in the pipe cross-section at  $Sc = 1000$ , from the parabolic marcher. Rows: Koch generation  $N = 0$ – $5$  (interfacial folding increases downward). Columns: streamwise station  $x = 0$  (inlet) to  $x = 18$  (outlet). The inlet interface ( $x = 0$ ) is the prescribed generation- $N$  Koch fold; downstream the fractal striations diffuse and the field homogenises faster at higher  $N$  (the per-panel  $M$  confirms the trend). Same 50/50 split and smooth wall for every  $N$ .

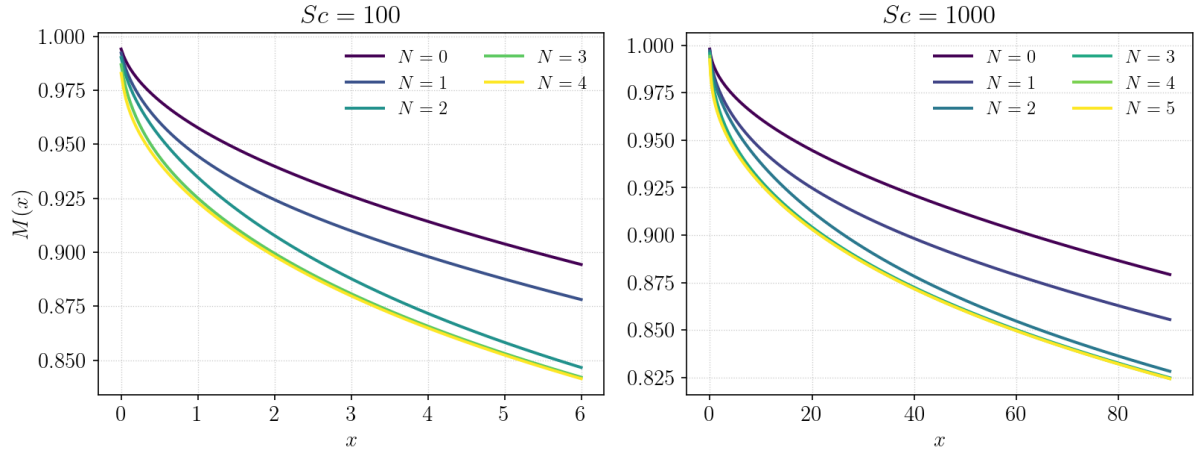


Figure 6: Segregation decay  $M(x)$ , generations  $N$  overlaid, for  $Sc = 100$  ( $L_x = 6$ , left) and  $Sc = 1000$  ( $L_x = 90$ , right). Higher  $N$  decays faster and further in both. Both cross  $M = 0.9$  within their domains, so the mixing length is read at the same threshold for every  $Sc$  (Table 1).